

# Evaluation Protocol and Filtering Order in Decision Tree–Naïve Bayes Hybrids

Ikramul Hasan Moral

Department of Computer Science and Engineering  
United International University  
Dhaka, Bangladesh  
imoral223489@bscse.uui.ac.bd

Dewan Md. Farid

Department of Computer Science and Engineering  
United International University  
Dhaka, Bangladesh  
dewanfarid@cse.uui.ac.bd

**Abstract**—We evaluate how supervised filtering order and the placement of cleaning relative to cross-validation affect decision tree–Naïve Bayes hybrids. The studied steps remove training observations misclassified by a Naïve Bayes judge and select attributes using a decision tree, with inverse-square-root depth weights for Naïve Bayes. We compose these steps in both orders and compare them with uncleaned classifiers, single-step methods, and an independent parallel composition. The evaluation uses Weka C4.5 and a mixed nominal/numeric Naïve Bayes implementation on ten benchmark datasets, with ten repetitions of stratified 10-fold cross-validation. Under training-fold-specific cleaning, the two combined tree methods average 81.21% and 80.89% accuracy, compared with 86.33% for uncleaned C4.5. The combined Naïve Bayes methods average 75.63% and 76.35%, compared with 75.93% without cleaning. Cleaning the complete dataset before cross-validation raises the combined tree scores to 94.72% and 96.32%, but also changes the population being evaluated. These differences therefore do not isolate leakage bias. Exploratory dataset-level tests provide no significant comparison after correction across 24 hypotheses. Removal records reveal class-dependent deletion and changes in tree-selected attributes after instance filtering. The findings support explicit evaluation and removal diagnostics for these hybrids; they do not establish improved classification or verified noise detection.

**Index Terms**—decision tree, Naïve Bayes, instance filtering, attribute weighting, cross-validation, evaluation protocol

## I. INTRODUCTION

Supervised cleaning can change both the observations available to a classifier and the attributes it uses. Farid et al. [1] introduced two independent decision tree (DT)–Naïve Bayes (NB) hybrids: an NB judge removes training observations before a tree is grown, and a tree selects and weights attributes before NB is fitted. These steps offer a direct setting in which to examine whether composing two cleaners adds value beyond their individual use.

The order of the steps changes their inputs. After instance removal, a tree selects attributes from a different training sample. After attribute selection, NB judges observations through a different representation. However, neither composition nor a smaller training set guarantees a better classifier. Earlier work on joint instance and feature selection already examined ordering [2], while research on misclassification-based filters identified the risk of rejecting valid exceptions [3], [4].

Evaluation introduces a separate issue. A supervised cleaner fitted before cross-validation can use information from future

held-out observations. This is related to established concerns about feature-selection leakage and performance evaluation [5]–[7]. Instance removal also changes the evaluation population if rejected rows disappear before folds are created. A higher score on those surviving rows answers a different question from a score on every held-out observation.

We investigate three questions. Does either composition improve on its matched uncleaned classifier or single-step alternatives? Does changing order alter classification and removal behavior? How do scores change when cleaning is performed before, rather than within, cross-validation? The last comparison is a protocol sensitivity analysis; it does not identify which implementation was used in the original study.

We implement the two steps on the original attribute columns with Weka J48 and a mixed-likelihood NB model. All main comparisons share training and test splits. Single-step and parallel reference methods help separate composition from the effects already present in individual steps. The main protocol refits supervised cleaning inside each training fold and scores every test row.

The study contributes an implementation-explicit comparison of the two orders, a descriptive contrast between two evaluation protocols, and fold-level removal diagnostics. Its contribution is empirical analysis of these particular hybrids, rather than a new tree-weighting rule or a claim that filtering generally fails. The tested combined methods do not show a consistent advantage over their uncleaned counterparts.

## II. RELATED WORK

### A. Combining instance and attribute operations

Farid et al. [1] used NB filtering for a tree and tree-derived weights for NB as separate algorithms. Hall [8] also derived NB attribute weights from minimum tree depth, but stabilized them through bagged unpruned trees. Thus the weighting principle is prior work; our implementation uses one pruned tree, following the studied single-tree design. Tsai et al. [2] investigated feature and instance selection priorities using genetic algorithms with other classifiers. Their study motivates an ordering comparison, but does not supply results for the specific NB/tree steps tested here. We make no claim to be the first study of preprocessing order.

### B. Disagreement is not a verified label error

Brodley and Friedl [3] used classifier-based filtering and examined the tradeoff between rejecting mislabeled and correctly labeled observations. Their filters obtained predictions through internal cross-validation, whereas our studied judge uses its training predictions. John [4] iteratively removed errors of a pruned tree and discussed potential losses on valid, underrepresented patterns. These findings motivate a distinction between classifier disagreement and known corruption. More recent work on confident learning [9] and the comparison of correction with filtering [10] addresses label uncertainty through different procedures. Those methods are context, not evaluated competitors in this study.

### C. Evaluation and statistical comparison

Cawley and Talbot [6] analyzed selection bias in performance evaluation, while Demircioğlu [5] examined feature selection before versus within cross-validation. Kapoor and Narayanan [7] situated leakage within broader reproducibility problems. We apply their methodological concern to supervised DT–NB cleaning, while explicitly recognizing that row filtering changes the cases subsequently scored. For statistical comparisons, we use dataset-level summaries following Demšar [11]; repetitions of overlapping folds are not treated as additional independent datasets.

## III. METHODS

### A. Base classifiers and notation

Let  $D$  contain labeled training observations and let  $S$  denote the selected attributes. DT is Weka 3.8.6 J48 (C4.5), with pruning enabled, confidence factor 0.25 and minimum leaf size 2 [12]. NB uses empirical class priors, add-one smoothing for nominal attributes, and a class-conditional Gaussian likelihood for numeric attributes. Numeric population variances receive  $10^{-9}$  times the largest numeric attribute variance in the current training sample. The implementation scores a class  $c$  by

$$\log P(c) + \sum_{j \in S} w_j \log P(x_j | c), \quad (1)$$

and predicts the class with the greatest score. Plain NB sets  $w_j = 1$ . A class absent from NB’s training sample cannot be predicted by that fit.

### B. Algorithm 1: instance filtering (N)

The NB judge is fitted to its supplied training sample and predicts those same rows. Rows whose predictions disagree with their labels are removed. This is in-sample disagreement filtering, not out-of-fold label-error estimation. If removal would leave fewer than two classes, the operation is skipped and all supplied rows are retained. The safeguard does not preserve every original class in a multiclass dataset. We call discarded rows *removed observations*, since no independent corruption labels are available.

The original single-step arrangement ends with DT and is denoted N–DT. N–NB is a reference using the same removal decision with NB as the final classifier. Neither method filters held-out rows under the main protocol.

**Require:** Training data  $D$ ; order  $o$ ; final classifier  $f$

**Ensure:** Fitted model  $M$  and selected attributes  $S$

```

1:  $S \leftarrow$  all attributes;  $W \leftarrow$  unit weights
2: if  $o = \text{AN}$  then
3:   Fit A on  $D$ ; obtain  $S, W$ 
4:   If  $S$  is empty, keep all attributes at unit weights
5:   Fit weighted NB on  $D[:, S]$ ; identify disagreements
6:   Remove disagreements if at least two classes remain
7: else
8:   Fit plain NB on  $D$ ; identify disagreements
9:   Remove disagreements if at least two classes remain
10:  Fit A on surviving  $D$ ; obtain  $S, W$ 
11:  If  $S$  is empty, keep all attributes at unit weights
12: end if
13: if  $f = \text{NB}$  then
14:   Fit NB on surviving  $D[:, S]$  with  $W$ 
15: else
16:   Fit J48 on surviving  $D[:, S]$ 
17: end if
18: return  $M, S$ 

```

Fig. 1. Algorithm 3: the combined hybrid fitted to one training fold. NA means instances first; AN means attributes first. Both safeguards are implementation choices. At prediction time, select  $S$  on every held-out row and apply  $M$ ; do not remove held-out rows.

### C. Algorithm 2: attribute selection and weighting (A)

A tree is grown on the supplied training rows. Attribute  $j$  receives  $w_j = 1/\sqrt{d_j}$ , where  $d_j$  is the smallest depth at which that tree tests it and root depth is one. Attributes not tested are omitted. A–NB uses these weights in Eq. 1; A–DT refits a tree on the retained attributes without using NB weights. Selection is based on original attributes, not separate one-hot indicators.

An empty-selection safeguard matters for reproducibility. In the main fold-specific implementation, an empty tree selection retains all attributes at unit weights, including for A–NB. In the before-CV implementation, standalone A–NB instead uses a priors-only model when selection is empty; composed paths retain all attributes. No empty-selection or class-collapse safeguard was triggered in the saved main-protocol records, so the empty-tree difference was not exercised there.

### D. Algorithm 3: ordered and parallel compositions

NA applies N using plain NB and then fits A on the surviving rows. AN fits A first and passes its selected attributes and weights to the NB judge in N. The latter weights also enter the final NB. Either order ends with NB or DT, giving four combined methods (Fig. 1).

A parallel reference, P, fits plain-NB filtering and tree attribute selection independently on the same original training fold. The final classifier uses the intersection of retained rows and retained attributes; final NB uses the tree’s weights. Comparing P with NA or AN measures the consequence of sequential dependence under these implementation choices. It does not isolate ordering from judge weighting: AN also changes the representation and weights used by its judge.

TABLE I  
DATASET DIMENSIONS IN THE FILES USED.

| Dataset            | Rows  | Attributes | Classes |
|--------------------|-------|------------|---------|
| Breast cancer      | 286   | 9          | 2       |
| Contact lenses     | 24    | 4          | 3       |
| Diabetes           | 768   | 8          | 2       |
| Glass              | 214   | 9          | 6       |
| Iris               | 150   | 4          | 3       |
| Soybean            | 683   | 35         | 19      |
| Vote               | 435   | 16         | 2       |
| Image segmentation | 2310  | 19         | 7       |
| Tic-tac-toe        | 958   | 9          | 2       |
| NSL-KDD            | 25192 | 41         | 22      |

#### IV. EXPERIMENTAL SETUP

##### A. Datasets and preprocessing

We use the ten dataset names considered by Farid et al., with the locally available files summarized in Table I. Iris is loaded through scikit-learn; NSL-KDD uses its 20% training subset [13]. Image segmentation combines the two available UCI files. Consequently its 2,310 rows differ from the 1,500 reported in the original study. Glass has six observed classes rather than seven; NSL-KDD has 22 rather than 23. These are implementation-specific benchmark experiments, not an exact reproduction of the original unpublished implementation.

Original nominal attributes remain single columns. Row identifiers and NSL-KDD’s difficulty field are excluded. Literal question-mark values are treated as nominal categories, without imputation. No numeric scaling is applied. Nominal vocabularies are constructed once from the complete files before splitting. They do not use labels, but influence NB smoothing denominators. Thus the main experiment is fold-local in supervised fitting, with a shared, label-free attribute schema; it is not a completely inductive preprocessing pipeline.

##### B. Two evaluation protocols

Protocol B, the main evaluation, uses ten repetitions of shuffled, stratified 10-fold cross-validation with seeds 0–9. Within a split, all 12 methods start with identical training and test rows. Every supervised cleaning step and final classifier is refitted using training rows only. A test fold loses the columns excluded by its training-fold selector but retains all its rows. Each method has 100 fold scores per dataset.

Protocol A fits the supervised cleaners once on the complete dataset, then cross-validates only the final classifier on the resulting rows and attributes using the same seed range. Filtering can use labels of future test cases and exclude those cases from evaluation altogether. Its accuracy therefore describes selected survivors, not held-out performance on the original population. Protocol-A folds are not paired with B folds. The A-minus-B difference combines information access, sample selection, and changes in fitted models; it is not an isolated causal estimate of leakage bias. We do not infer that the original study used A.

Ten folds are retained even when classes have fewer than ten observations. Contact lenses has class counts 4, 5, and 15, and NSL-KDD includes singleton classes. Some training folds therefore lack classes found in their test folds. This limitation is shared by the compared methods and must be considered when interpreting class-sensitive measures.

##### C. Measures, aggregation, and exploratory inference

Accuracy and macro-F1 are calculated for each held-out fold. Macro-F1 averages classes occurring in that fold’s true labels or predictions, assigning zero F1 where necessary; classes absent from both are excluded. The contributing class set can therefore vary across folds. Dataset scores are arithmetic means of 100 fold percentages, rather than metrics computed from pooled predictions. Overall means weight the ten datasets equally. We report the sample standard deviation of the overall mean across the ten split seeds as descriptive split sensitivity, not as a standard error or confidence interval.

For each cleaning event we record rows and attributes before and after cleaning, plus the removal fraction within each class. Event totals are counted once per dataset, seed, fold, and stage; per-class records must not be counted as additional cleaning events.

The statistical analysis was added after inspecting the saved results and is exploratory. Twelve contrasts cover the original single-step methods against their matched baselines, four combined methods against baselines, two order comparisons, and four combined-versus-parallel comparisons. Each is tested for accuracy and macro-F1, making a family of 24 hypotheses. Two-sided Wilcoxon signed-rank tests use ten paired dataset means, with zero differences excluded and exhaustive sign permutations to handle ties. Differences are rounded to ten decimal places before ranking. Holm correction is applied jointly across the 24 tests at level 0.05. Neither 100 overlapping folds nor ten repetitions increase the number of independent datasets [11]. Protocol contrasts and removal diagnostics are descriptive and are not included in this family.

#### V. RESULTS

##### A. Classification under fold-specific cleaning

Table II summarizes all 12 methods, and Tables III and IV retain their per-dataset results. N and A denote the instance and attribute steps; NA and AN denote the orders; P denotes independent parallel composition. N–DT and A–NB are the original single-step arrangements, while N–NB and A–DT are attribution references.

Uncleaned C4.5 achieved 86.33% mean accuracy and 81.92% macro-F1. NA–DT and AN–DT achieved 81.21% and 80.89% accuracy, respectively: differences of –5.12 and –5.44 percentage points (pp). Their macro-F1 scores were lower by 6.16 and 7.99 pp. A–DT remained close to its baseline at 86.19% accuracy, whereas N–DT averaged 81.03%. Thus adding attribute selection alone changed the tree mean little; methods involving instance removal had lower means on this suite.

TABLE II  
 PROTOCOL-B EQUAL-DATASET MEANS AND SPLIT-SEED STANDARD  
 DEVIATIONS (IN PERCENTAGE POINTS). HIGHER SCORES ARE BETTER.  
 BOLD MARKS THE HIGHEST MEAN, WITHOUT IMPLYING A SIGNIFICANT  
 DIFFERENCE.

| Method | Accuracy (%)                       | Macro-F1 (%)                       |
|--------|------------------------------------|------------------------------------|
| NB     | 75.93 $\pm$ 0.32                   | 69.80 $\pm$ 0.52                   |
| C4.5   | <b>86.33 <math>\pm</math> 0.33</b> | <b>81.92 <math>\pm</math> 0.44</b> |
| N-NB   | 75.42 $\pm$ 0.46                   | 69.08 $\pm$ 0.46                   |
| N-DT   | 81.03 $\pm$ 0.37                   | 75.70 $\pm$ 0.63                   |
| A-NB   | 77.35 $\pm$ 0.41                   | 71.73 $\pm$ 0.63                   |
| A-DT   | 86.19 $\pm$ 0.33                   | 81.84 $\pm$ 0.42                   |
| NA-NB  | 75.63 $\pm$ 0.45                   | 69.60 $\pm$ 0.72                   |
| NA-DT  | 81.21 $\pm$ 0.29                   | 75.76 $\pm$ 0.58                   |
| AN-NB  | 76.35 $\pm$ 0.34                   | 69.91 $\pm$ 0.60                   |
| AN-DT  | 80.89 $\pm$ 0.45                   | 73.93 $\pm$ 0.80                   |
| P-NB   | 76.39 $\pm$ 0.37                   | 70.47 $\pm$ 0.68                   |
| P-DT   | 81.05 $\pm$ 0.34                   | 75.58 $\pm$ 0.58                   |

For NB, A-NB averaged 77.35% accuracy and 71.73% macro-F1, compared with 75.93% and 69.80% without cleaning. The combined NA-NB and AN-NB accuracies were 75.63% and 76.35%; their baseline differences were  $-0.30$  and  $+0.42$  pp. AN-NB had higher accuracy than NA-NB on eight of ten datasets, but its mean advantage was only 0.71 pp. The tree order comparison went in the opposite direction on the overall mean: AN-DT was 0.32 pp below NA-DT. There is no common winning order across final classifiers and metrics.

Dataset results expose substantial heterogeneity. On glass, C4.5 scored 66.58% accuracy, compared with 48.97% for NA-DT and 49.15% for AN-DT. On vote, the corresponding scores were 95.08%, 95.13%, and 95.54%. For NB on glass, the baseline scored 45.95%, whereas the combined methods scored 30.93% and 37.28%. These contrasts show why an overall mean alone is insufficient to characterize the cleaners.

#### B. Sequential dependence and statistical uncertainty

P-NB averaged 76.39% accuracy and 70.47% macro-F1; P-DT averaged 81.05% and 75.58%. The sequential methods did not show a consistent mean advantage over this independent reference. Table V reports both raw and Holm-adjusted  $p$ -values for the full contrast family. None met the corrected 0.05 threshold. AN-DT versus C4.5 had raw accuracy  $p = 0.014$ , but adjusted  $p = 0.328$ . The NB order comparison had raw  $p = 0.275$ , and the DT order comparison had raw  $p = 0.922$ . These tests do not establish equality or practical equivalence. With ten heterogeneous datasets, a nonsignificant result can coexist with meaningful losses on particular tasks.

#### C. Before-CV cleaning changes the apparent result

The protocol contrast in Table VI was much larger for methods involving instance removal than for attribute selection alone. NA-DT increased from 81.21% in B to 94.72% in A, and AN-DT from 80.89% to 96.32%, differences of 13.51 and 15.43 pp. N-DT differed by 13.50 pp. In contrast, A-NB differed by 0.72 pp and A-DT by 0.33 pp. These are

observed score differences under different procedures and, for row-filtering methods, different evaluation samples. They are not estimates of deployable improvement.

#### D. What the filters removed

Table VII summarizes 100 events per stage per dataset. Plain-NB removal ranged from 4.14% of training rows on iris to 47.31% on glass. The tree retained 97.00% of glass attributes when fitted on the original training fold, but 61.00% after plain-NB filtering. On vote, these fractions were 27.38% and 7.62%. These observations directly show that the first operation changes the fitted output of the second, even where classifier scores are close.

Breast-cancer recurrence cases made up 29.72% of training rows on average. Plain NB removed 50.63% of that class versus 14.51% of non-recurrence cases. After attribute selection, weighted NB removed 58.97% and 11.72%, respectively. In contact lenses, the weighted filter entirely removed class 1 from 12 of 100 training folds; in glass, it entirely removed class 3 from nine folds. The two-class safeguard did not prevent these losses because other classes remained. These are class-removal diagnostics, not evidence that the removed labels were wrong.

### VI. DISCUSSION

The reference methods narrow the interpretation of composition. Attribute selection followed by DT produced almost the baseline tree mean, while adding instance filtering did not recover an overall tree advantage. For NB, the attribute-only method had a higher mean than either combined order. Composition therefore cannot be justified from the premise that both component steps are beneficial independently. Their outputs interact, but an interaction in retained rows or attributes need not improve held-out prediction.

The class-removal records are consistent with the concern raised by Brodley and Friedl [3] and John [4]: a classifier can reject difficult but valid observations. Our results do not establish that mechanism as the cause of the losses, since we lack verified noise labels and a controlled corruption intervention. Nor do they contradict benefits reported for different label-cleaning approaches [9], [10]. The warranted conclusion concerns this in-sample NB disagreement rule and these benchmark conditions.

Protocol A reveals how an apparently strong score can coexist with a weaker evaluation on all held-out cases. It combines access to future test information with selection of cases that survive a supervised judge. The distinction is especially consequential when filtering removes a whole class. Reporting only the survivor score would conceal that change in the prediction task. A useful follow-up would keep the original test population fixed while varying access to training information, separating selection effects from leakage more directly.

### VII. LIMITATIONS AND REPRODUCIBILITY

These ten legacy datasets were chosen for continuity with the seed study, not as a random sample of classification tasks.

TABLE III  
 PROTOCOL-B ACCURACY (%) PER DATASET. NB-FINAL METHODS ARE IN THE UPPER PANEL; DT-FINAL METHODS ARE IN THE LOWER PANEL.

| Dataset            | NB    | N-NB  | A-NB  | NA-NB | AN-NB | P-NB  |
|--------------------|-------|-------|-------|-------|-------|-------|
| breast-cancer      | 72.12 | 72.68 | 72.43 | 71.28 | 72.53 | 72.46 |
| contact-lenses     | 73.00 | 73.00 | 80.67 | 83.00 | 79.33 | 83.00 |
| diabetes           | 75.52 | 74.41 | 76.73 | 75.58 | 75.67 | 75.81 |
| glass              | 45.95 | 37.00 | 47.84 | 30.93 | 37.28 | 32.29 |
| image-segmentation | 79.76 | 74.92 | 82.30 | 74.53 | 77.32 | 76.57 |
| iris               | 95.53 | 95.33 | 95.40 | 95.33 | 95.87 | 95.60 |
| nsl-kdd            | 67.42 | 75.79 | 62.80 | 74.53 | 75.23 | 76.16 |
| soybean            | 90.09 | 89.00 | 89.40 | 85.67 | 86.66 | 86.95 |
| tic-tac-toe        | 69.83 | 71.89 | 70.68 | 70.03 | 68.07 | 69.72 |
| vote               | 90.06 | 90.13 | 95.24 | 95.45 | 95.49 | 95.36 |

  

| Dataset            | C4.5  | N-DT  | A-DT  | NA-DT | AN-DT | P-DT  |
|--------------------|-------|-------|-------|-------|-------|-------|
| breast-cancer      | 74.84 | 71.77 | 73.91 | 71.77 | 72.08 | 71.20 |
| contact-lenses     | 84.67 | 84.67 | 84.67 | 84.67 | 80.67 | 84.67 |
| diabetes           | 74.12 | 75.12 | 74.18 | 75.23 | 75.57 | 74.86 |
| glass              | 66.58 | 48.83 | 66.58 | 48.97 | 49.15 | 48.65 |
| image-segmentation | 96.81 | 89.24 | 96.83 | 89.26 | 90.88 | 89.79 |
| iris               | 94.27 | 94.93 | 94.27 | 94.93 | 93.80 | 94.93 |
| nsl-kdd            | 99.47 | 87.43 | 99.47 | 89.68 | 87.07 | 88.37 |
| soybean            | 92.71 | 88.71 | 92.26 | 87.89 | 90.98 | 88.24 |
| tic-tac-toe        | 84.74 | 74.43 | 84.74 | 74.55 | 73.17 | 74.43 |
| vote               | 95.08 | 95.13 | 95.03 | 95.13 | 95.54 | 95.38 |

Dataset dimensions, question-mark handling, NB variance stabilization, and tree settings can differ from the original implementation. We did not reproduce its complete software or establish its cleaning placement. In particular, the original study’s aggregate claims cannot be attributed to leakage from this audit.

Rare classes and small folds limit macro-F1 interpretation. A variable fold class set is different from averaging a fixed dataset-wide class set, and NB cannot predict an absent training class. The shared full-file nominal vocabulary is another boundary on inductive evaluation. No controlled label-noise experiment, modern classifier comparison, external NSL-KDD test-set evaluation, or runtime benchmark is included. The exploratory tests account for multiple comparisons but do not turn this selected benchmark suite into a population-level sample.

The reported results derive from saved metrics, per-fold scores, removal records, and configuration files. The table-generation script recomputes dataset means, checks them against the stored aggregates, verifies the expected fold coverage, and derives the statistical comparisons and removal tables. The numerical audit uses the saved experiments; it does not constitute a fresh rerun of every model. The analysis can be reproduced from the repository root by running `python paper/manuscript/scripts/make_results_table.py`.

## VIII. CONCLUSION

Composing NB disagreement filtering with tree-based attribute selection changed removal behavior without producing a consistent classification advantage on the tested suite. The

observed order preference depended on the final classifier, and no exploratory comparison remained significant after correction. Cleaning before cross-validation yielded much higher scores for row-filtering methods, while also changing which observations were evaluated. Together, these findings motivate reporting the cleaning protocol, matched baselines, and per-class removal records alongside accuracy. Future work should test the suspected hard-case mechanism with known corruption labels, preserve rare classes explicitly, and compare protocols on a fixed held-out population.

## REFERENCES

- [1] D. Md. Farid, L. Zhang, C. M. Rahman, M. A. Hossain, and R. Strachan, “Hybrid decision tree and naïve Bayes classifiers for multi-class classification tasks,” *Expert Systems with Applications*, vol. 41, no. 4, pp. 1937–1946, 2014.
- [2] C.-F. Tsai, W. Eberle, and C.-Y. Chu, “Genetic algorithms in feature and instance selection,” *Knowledge-Based Systems*, vol. 39, pp. 240–247, 2013.
- [3] C. E. Brodley and M. A. Friedl, “Identifying mislabeled training data,” *Journal of Artificial Intelligence Research*, vol. 11, pp. 131–167, 1999.
- [4] G. H. John, “Robust decision trees: Removing outliers from databases,” in *Proc. 1st Int. Conf. Knowledge Discovery and Data Mining*, 1995, pp. 174–179.
- [5] A. Demircioğlu, “Measuring the bias of incorrect application of feature selection when using cross-validation in radiomics,” *Insights into Imaging*, vol. 12, art. 172, 2021.
- [6] G. C. Cawley and N. L. C. Talbot, “On over-fitting in model selection and subsequent selection bias in performance evaluation,” *Journal of Machine Learning Research*, vol. 11, pp. 2079–2107, 2010.
- [7] S. Kapoor and A. Narayanan, “Leakage and the reproducibility crisis in machine-learning-based science,” *Patterns*, vol. 4, no. 9, art. 100804, 2023.
- [8] M. Hall, “A decision tree-based attribute weighting filter for naïve Bayes,” *Knowledge-Based Systems*, vol. 20, no. 2, pp. 120–126, 2007.

TABLE IV  
PROTOCOL-B MACRO-F1 (%) PER DATASET, WITH THE SAME METHOD PANELS AS TABLE III. FOLD CLASS SETS FOLLOW SECTION IV-C.

| Dataset            | NB    | N-NB  | A-NB  | NA-NB | AN-NB | P-NB  |
|--------------------|-------|-------|-------|-------|-------|-------|
| breast-cancer      | 64.42 | 64.22 | 62.57 | 61.33 | 61.57 | 62.94 |
| contact-lenses     | 62.31 | 62.31 | 72.27 | 76.47 | 69.87 | 76.47 |
| diabetes           | 72.32 | 71.43 | 72.76 | 71.96 | 71.05 | 72.03 |
| glass              | 44.98 | 40.72 | 44.71 | 36.78 | 40.42 | 37.99 |
| image-segmentation | 77.78 | 74.52 | 81.42 | 73.40 | 77.06 | 75.82 |
| iris               | 95.48 | 95.28 | 95.35 | 95.29 | 95.82 | 95.56 |
| nsl-kdd            | 34.06 | 35.87 | 37.58 | 35.36 | 38.18 | 36.46 |
| soybean            | 93.13 | 92.59 | 91.33 | 87.27 | 88.40 | 89.95 |
| tic-tac-toe        | 63.83 | 64.08 | 64.26 | 62.93 | 61.45 | 62.30 |
| vote               | 89.66 | 89.80 | 95.02 | 95.25 | 95.29 | 95.15 |

  

| Dataset            | C4.5  | N-DT  | A-DT  | NA-DT | AN-DT | P-DT  |
|--------------------|-------|-------|-------|-------|-------|-------|
| breast-cancer      | 61.81 | 62.68 | 60.52 | 62.65 | 61.37 | 61.82 |
| contact-lenses     | 79.47 | 79.47 | 79.47 | 79.47 | 72.27 | 79.47 |
| diabetes           | 70.71 | 71.45 | 70.79 | 71.56 | 71.19 | 70.97 |
| glass              | 62.00 | 45.29 | 62.00 | 45.45 | 43.35 | 45.03 |
| image-segmentation | 96.81 | 88.96 | 96.83 | 88.98 | 90.79 | 89.53 |
| iris               | 94.18 | 94.88 | 94.18 | 94.88 | 93.72 | 94.88 |
| nsl-kdd            | 82.25 | 60.33 | 82.92 | 62.05 | 53.18 | 60.82 |
| soybean            | 94.26 | 92.08 | 94.10 | 90.47 | 92.08 | 91.21 |
| tic-tac-toe        | 82.82 | 66.94 | 82.82 | 67.18 | 66.02 | 66.94 |
| vote               | 94.86 | 94.90 | 94.81 | 94.90 | 95.34 | 95.17 |

TABLE V  
EXPLORATORY TWO-SIDED DATASET-LEVEL SIGNED-RANK COMPARISONS. DELTAS ARE FIRST METHOD MINUS SECOND IN PP;  $p_H$  IS HOLM-ADJUSTED ACROSS ALL 24 ACCURACY/F1 TESTS. F1 DENOTES MACRO-F1. ALL TESTS USE TEN DATASETS.

| Comparison     | $\Delta\text{Acc}$ | $p$   | $p_H$ | $\Delta\text{F1}$ | $p$   | $p_H$ |
|----------------|--------------------|-------|-------|-------------------|-------|-------|
| N-DT vs DT     | -5.30              | 0.055 | 1.000 | -6.22             | 0.164 | 1.000 |
| A-NB vs NB     | +1.42              | 0.131 | 1.000 | +1.93             | 0.193 | 1.000 |
| NA-NB vs NB    | -0.30              | 1.000 | 1.000 | -0.19             | 0.557 | 1.000 |
| AN-NB vs NB    | +0.42              | 0.846 | 1.000 | +0.11             | 0.922 | 1.000 |
| NA-DT vs DT    | -5.12              | 0.055 | 1.000 | -6.16             | 0.164 | 1.000 |
| AN-DT vs DT    | -5.44              | 0.014 | 0.328 | -7.99             | 0.037 | 0.854 |
| AN-NB vs NA-NB | +0.71              | 0.275 | 1.000 | +0.31             | 0.492 | 1.000 |
| AN-DT vs NA-DT | -0.32              | 0.922 | 1.000 | -1.83             | 0.232 | 1.000 |
| NA-NB vs P-NB  | -0.76              | 0.039 | 0.859 | -0.86             | 0.055 | 1.000 |
| AN-NB vs P-NB  | -0.05              | 0.770 | 1.000 | -0.56             | 0.846 | 1.000 |
| NA-DT vs P-DT  | +0.16              | 0.461 | 1.000 | +0.18             | 0.461 | 1.000 |
| AN-DT vs P-DT  | -0.16              | 0.770 | 1.000 | -1.65             | 0.193 | 1.000 |

- [9] C. G. Northcutt, L. Jiang, and I. L. Chuang, “Confident learning: Estimating uncertainty in dataset labels,” *Journal of Artificial Intelligence Research*, vol. 70, pp. 1373–1411, 2021.
- [10] G. Jiang, J. Zhang, X. Bai, W. Wang, and D. Meng, “Which is more effective in label noise cleaning, correction or filtering?” in *Proc. AAAI Conf. Artificial Intelligence*, vol. 38, no. 11, 2024, pp. 12866–12873.
- [11] J. Demšar, “Statistical comparisons of classifiers over multiple data sets,” *Journal of Machine Learning Research*, vol. 7, pp. 1–30, 2006.
- [12] I. H. Witten, E. Frank, M. A. Hall, and C. J. Pal, *Data Mining: Practical Machine Learning Tools and Techniques*, 4th ed. Morgan Kaufmann, 2016.
- [13] M. Tavallaei, E. Bagheri, W. Lu, and A. A. Ghorbani, “A detailed analysis of the KDD CUP 99 data set,” in *Proc. IEEE Symp. Computational Intelligence for Security and Defense Applications*, 2009, pp. 1–6.

TABLE VI  
MEAN ACCURACY UNDER B (TRAINING-FOLD CLEANING) AND A (FULL-DATA CLEANING BEFORE CV). DIFFERENCES ARE DESCRIPTIVE PROTOCOL CONTRASTS; A MAY EXCLUDE FUTURE TEST ROWS.

| Method | B (%) | A (%) | A–B (pp) |
|--------|-------|-------|----------|
| N–NB   | 75.42 | 91.93 | +16.51   |
| N–DT   | 81.03 | 94.52 | +13.50   |
| A–NB   | 77.35 | 78.07 | +0.72    |
| A–DT   | 86.19 | 86.52 | +0.33    |
| NA–NB  | 75.63 | 90.11 | +14.48   |
| NA–DT  | 81.21 | 94.72 | +13.51   |
| AN–NB  | 76.35 | 91.08 | +14.74   |
| AN–DT  | 80.89 | 96.32 | +15.43   |
| P–NB   | 76.39 | 91.10 | +14.71   |
| P–DT   | 81.05 | 94.67 | +13.62   |

TABLE VII  
MEAN REMOVAL/RETENTION FRACTIONS (%) UNDER B. N DEL. IS  
PLAIN-NB ROW DELETION; AN DEL. IS WEIGHTED-NB DELETION AFTER  
A. A KEPT AND NA KEPT ARE ATTRIBUTE RETENTION BEFORE AND  
AFTER PLAIN-NB FILTERING.

| Dataset            | N del. | AN del. | A kept | NA kept |
|--------------------|--------|---------|--------|---------|
| breast-cancer      | 25.24  | 25.76   | 41.56  | 34.67   |
| contact-lenses     | 4.28   | 11.03   | 66.50  | 64.00   |
| diabetes           | 23.67  | 22.99   | 84.88  | 83.12   |
| glass              | 47.31  | 46.88   | 97.00  | 61.00   |
| image-segmentation | 19.88  | 17.35   | 65.79  | 40.16   |
| iris               | 4.14   | 4.41    | 50.75  | 37.75   |
| nsf-kdd            | 32.52  | 37.13   | 69.32  | 42.59   |
| soybean            | 9.18   | 9.06    | 52.40  | 41.89   |
| tic-tac-toe        | 29.32  | 28.63   | 100.00 | 76.67   |
| vote               | 9.33   | 4.46    | 27.38  | 7.62    |